Part IX — Modern Physics and the Universe · Chapter 52

Stars, Galaxies, and Cosmology

How physics explains the birth, life, and large-scale structure of the universe

> *Every atom of iron in your blood was forged inside a star that exploded before the Solar System formed. The calcium in your bones, the oxygen you breathe, the carbon in your cells — all of it was synthesised in stellar interiors or in the violent deaths of massive stars. The universe began as a hot, dense plasma of quarks and leptons some 13.8 billion years ago, expanded and cooled, formed atoms, then clouds of gas, then stars, then galaxies, and eventually planets and organisms capable of wondering about their origins. Physics — the same mechanics, thermodynamics, nuclear physics, electromagnetism, and relativity studied throughout this course — explains this entire arc from the Big Bang to the present structure of the cosmos. This capstone chapter of Part IX traces that story, showing how the physical laws you know scale up from the atomic nucleus to the observable universe.*

52.1 Physics at Cosmic Scales

The same physical laws that describe a ball rolling down a ramp or an electron jumping between energy levels also govern the motion of galaxies, the fusion reactions inside stars, and the expansion of the universe itself. This universality is one of the most powerful results in all of science.

Cosmic scales are almost incomprehensibly large. The Sun is 1.5×10^{11} m from Earth (1 astronomical unit, AU). The nearest star, Proxima Centauri, is about 4.24 light-years (4.0×10^{16} m) away. The Milky Way galaxy is about 100 000 light-years across. The observable universe extends about 46 billion light-years in every direction. Yet the equations of gravity, nuclear physics, and thermodynamics apply equally to all of these scales.

Scale	Distance or Size	Comparison
Earth radius	6.37×10^6 m	Reference
Earth–Moon	3.84×10^8 m	60 Earth radii

Earth–Sun (1 AU)	1.50×10^{11} m	23 500 Earth radii
Nearest star (Proxima Cen)	4.0×10^{16} m (4.24 ly)	270 000 AU
Milky Way diameter	~ 10^{2}± m (~ 10^{5} ly)	30 kpc
Local Group diameter	~ 3×10^{22} m (~ 3 Mly)	1 Mpc
Observable universe radius	~ 4×10^{26} m (46 Gly)	14 Gpc

Table 52.1 *Characteristic scales of cosmic structures. ly = light-year; pc = parsec (3.26 ly); kpc = kiloparsec; Mpc = megaparsec; Gpc = gigaparsec.*

52.2 Gravity as the Organizing Force of the Universe

Of the four fundamental forces — gravity, electromagnetism, the strong nuclear force, and the weak nuclear force — only gravity operates significantly at cosmic scales. The other forces are either short-ranged (strong and weak nuclear) or cancel out at large scales because matter is electrically neutral on large scales.

Gravity is the weakest of the four forces at the subatomic level: the gravitational attraction between two protons is about 10^{36} times weaker than their electromagnetic repulsion. Yet gravity wins at cosmic scales for three reasons: (1) it is always attractive, never repulsive; (2) it is long-range, falling only as $1/r^2$; and (3) mass is always positive and cumulative — adding more mass always adds more gravity.

Gravity structures the universe at every scale. It pulls gas clouds together to form stars. It holds stars together in galaxies. It clusters galaxies into groups and superclusters. On the very largest scales ($\gtrsim$1 Gpc), the distribution of matter in the universe forms a web -like structure of filaments, sheets, and voids — the cosmic web — sculpted entirely by gravity acting on matter since the Big Bang.

The escape velocity from a gravitational body — the minimum speed needed to escape its gravity permanently — is given by $v\_esc = \sqrt{(2GM/r)}$. For Earth, this is about 11.2 km/s. For the Sun, about 620 km/s. For a neutron star, a significant fraction of the speed of light. For a black hole, the escape velocity from inside the event horizon exceeds c, so not even light can escape.

52.3 Star Formation

Stars form from clouds of gas and dust called nebulae — enormous regions of hydrogen, helium, and small amounts of heavier elements spread through the interstellar medium. A typical star-forming molecular cloud contains thousands to millions of solar masses of material and is extremely cold (~ 10–30 K) and dense by interstellar standards.

Star formation begins when a region of a nebula becomes gravitationally unstable — its own gravity overcomes the internal gas pressure and thermal motion that would otherwise prevent collapse. This can be triggered by the shockwave from a nearby supernova, the passage of a density wave in a galaxy's spiral arm, or simply by the accumulation of enough mass. Once collapse begins, it accelerates: as the cloud contracts, its gravity strengthens, pulling material inward faster.

As the cloud collapses, it heats up (gravitational potential energy converts to kinetic energy, which converts to thermal energy). The collapsing centre becomes a protostar — a hot, dense object still accreting material. The protostar continues to shrink and heat until its core temperature reaches about 10 million kelvin, at which point hydrogen fusion ignites. The outward pressure of the fusion radiation balances the inward pull of gravity: the protostar becomes a main-sequence star. For a star like the Sun, this process takes about 50 million years.

> **Key Point:** The Jeans criterion determines when a gas cloud is unstable to gravitational collapse: roughly, collapse occurs when the cloud's gravitational potential energy exceeds its thermal kinetic energy. Larger, colder, denser clouds are more susceptible to collapse. The Jeans mass for a molecular cloud at 10 K and typical density is a few solar masses — consistent with observed star masses.

52.4 Nuclear Fusion in Stars

The energy source of stars is nuclear fusion (Chapter 50). In main-sequence stars like the Sun, the dominant process is the proton-proton chain: four hydrogen nuclei (protons) are fused, through a series of steps, into one helium-4 nucleus. The helium nucleus has slightly less mass than the four protons. The mass difference is converted to energy via $E = \Delta mc^2$.

$$4\ {}^{1}\mathrm{H} \rightarrow {}^{4}\mathrm{He} + 2e^{+} + 2\nu_{e} + \text{energy}$$

The proton-proton chain (summarised). Four hydrogen nuclei fuse to form one helium nucleus, releasing two positrons, two neutrinos, and about 26.7 MeV of energy. The mass of helium is about 0.7% less than the total mass of four protons; this 0.7% is the source of all solar energy.

The Sun converts approximately 600 million tonnes of hydrogen to helium every second, releasing 3.85×10^{26} W of power. This energy takes about 100 000 years to diffuse from the core to the surface, where it is radiated into space. The light reaching Earth left the Sun's surface about 8 minutes ago, but originated in a fusion reaction roughly 100 000 years ago.

In more massive and hotter stars, the CNO cycle dominates: carbon, nitrogen, and oxygen act as catalysts in fusing hydrogen into helium. More massive stars can also fuse helium into carbon (triple-alpha process), carbon into neon and magnesium, and so on up the periodic table. Elements up to iron (Fe, atomic number 26) can be produced by fusion because fusing elements lighter than iron releases energy. Fusing elements heavier than iron requires energy — these elements are produced only in supernova explosions.

The balance between radiation pressure (outward) and gravity (inward) determines a star's structure and lifespan. This hydrostatic equilibrium is stable as long as fuel is available. When the fuel runs out, the balance breaks down and the star evolves rapidly.

> **Historical Note:** The source of stellar energy was a major unsolved problem until the 1930s. In 1920, Arthur Eddington speculated that stars were powered by the fusion of hydrogen into helium. The detailed mechanism — the proton-proton chain — was worked out by Hans Bethe in 1938. Bethe received the Nobel Prize in Physics in 1967. Fred Hoyle's work in the 1950s showed how elements heavier than helium are synthesised in stars and supernovae — a process called stellar nucleosynthesis.

52.5 Stellar Life Cycles

A star's fate is determined almost entirely by its initial mass. More massive stars burn hotter and faster, live shorter lives, and die more violently. The Sun, a medium-mass star, will live for about 10 billion years total (it is currently about 4.6 billion years old).

Low-mass stars (< about 8 solar masses)

These stars fuse hydrogen in their cores for billions to tens of billions of years. When the hydrogen is exhausted, the core contracts and heats while the outer layers expand and cool, forming a red giant. Helium fusion then ignites in the core (triple-alpha process). Eventually, the outer layers are expelled as a planetary nebula, leaving behind a white dwarf: a dense, Earth-sized remnant of the carbon-oxygen core, no longer producing energy by fusion, slowly cooling over billions of years.

High-mass stars (> about 8 solar masses)

Massive stars live only millions of years (compared to billions for the Sun). They are much brighter and hotter (blue or blue-white in colour). They fuse progressively heavier elements in their cores: hydrogen → helium → carbon → neon → oxygen → silicon → iron. Each stage is shorter than the last (silicon burning lasts only a few days). When an iron core forms, no more energy can be extracted by fusion. The core collapses catastrophically in less than a second, producing a supernova explosion.

Star Mass (solar masses)	Main-Sequence Lifespan	Colour	Final Stage
0.1	~ 10 trillion years	Red (M-type)	White dwarf
0.5	~ 56 billion years	Orange-red (K-type)	White dwarf
1.0 (Sun)	~ 10 billion years	Yellow (G-type)	White dwarf
5.0	~ 70 million years	White (A-type)	White dwarf
15	~ 11 million years	Blue-white (B-type)	Neutron star
40	~ 1 million years	Blue (O-type)	Black hole

Table 52.2 *Stellar properties and fates by mass. More massive stars are hotter, shorter-lived, and leave denser remnants.*

Bridge Note *The life cycle of stars connects directly to the nuclear physics of Chapters 49–50: fusion, fission, and radioactive decay all operate inside stellar interiors and remnants. The chemical elements that make up planets, oceans, and living organisms were assembled in stellar interiors over billions of years before the Solar System formed.*

52.6 Supernovae, Neutron Stars, and Black Holes

When a massive star's iron core collapses, the collapse is so rapid and energetic that it triggers a Type II supernova: one of the most violent events in the universe. The explosion releases in a few seconds as much energy as the Sun will emit over its entire 10-billion-year lifetime. For a brief period, a supernova can outshine its entire host galaxy.

Supernovae perform two crucial functions. First, they synthesise and expel elements heavier than iron (gold, uranium, platinum, etc.) — these elements are produced by rapid neutron capture (the r-process) during the explosion. Second, they blow these newly synthesised heavy elements into the interstellar medium, enriching it with the raw materials for future stars, planets, and eventually biology. Every gold atom on Earth was made in a supernova or neutron star merger.

Neutron Stars

If the original star's mass was between about 8 and 20 solar masses, the collapsed core may be stabilised by neutron degeneracy pressure (a quantum mechanical effect, similar to electron degeneracy pressure in white dwarfs but stronger). The result is a neutron star: an object about 20 km in diameter with a mass of 1.4–2.5 solar masses. Its density is approximately nuclear density — about 10^{17} kg/m^3, or 400 million tonnes per cubic centimetre. A rapidly spinning neutron star emitting beams of radio waves is called a pulsar.

Black Holes

If the collapsing core exceeds the maximum neutron star mass (the Tolman–Oppenheimer–Volkoff limit, about 2–3 solar masses), nothing can halt the collapse. The core collapses to a point of infinite density called a singularity, surrounded by an event horizon: the surface within which the escape velocity exceeds c. This is a black hole. Nothing — not matter, not light, not information — can escape from within the event horizon. Stellar-mass black holes have masses of 3–50 solar masses; supermassive black holes at the centres of galaxies can have masses of millions to billions of solar masses.

> **Key Point:** The existence of supermassive black holes at the centres of most large galaxies is now well-established. The black hole at the centre of the Milky Way, Sagittarius A*, has a mass of about 4 million solar masses. The Event Horizon Telescope collaboration produced the first image of a black hole shadow (in M87 galaxy) in 2019, and the first image of Sagittarius A* in 2022.

52.7 Galaxies and Galactic Structure

A galaxy is a gravitationally bound system containing stars, gas, dust, dark matter, and (usually) a supermassive black hole at its centre. Galaxies are the fundamental structural unit of the universe at intermediate scales — larger than star clusters but smaller than galaxy clusters.

Galaxies are classified by shape into four main types. Spiral galaxies (like the Milky Way and Andromeda) have a central bulge surrounded by a flat disc with spiral arms of younger, hotter stars and gas. Elliptical galaxies are spheroidal or ellipsoidal, contain mostly old stars and little gas, and range from dwarf ellipticals to giant ellipticals containing trillions of stars. Lenticular galaxies are a hybrid: disc-shaped like spirals but lacking prominent spiral arms. Irregular galaxies have no regular shape, often the result of gravitational interactions or mergers.

The distribution of galaxies is not random. Gravity has organized them into groups (a few to dozens of galaxies), clusters (hundreds to thousands of galaxies), superclusters (collections of clusters spanning hundreds of millions of light-years), and the cosmic web of filaments, walls, and voids that forms the largest-scale structure of the universe.

Galaxy Type	Shape	Typical Content	Star Formation
Spiral	Disc + bulge + arms	Young and old stars, gas, dust	Active in arms
Elliptical	Spheroidal	Mostly old red stars, little gas	Little to none
Lenticular	Disc without spiral arms	Old stars, minimal gas	Little to none
Irregular	No regular shape	Mixed stellar ages, gas rich	Often vigorous

Table 52.3 *Galaxy morphological types and their properties.*

52.8 The Milky Way

Our galaxy, the Milky Way, is a barred spiral galaxy (a spiral galaxy with a bar-shaped structure through the central bulge) approximately 100 000 light-years in diameter and about 1000 light-years thick at the disc. It contains an estimated 200–400 billion stars and a total mass (including dark matter) of about 1.5×10^{12} solar masses.

The Solar System is located about 26 000 light-years from the galactic centre — roughly halfway out from the centre to the edge — in a minor spiral arm called the Orion Spur, between the Perseus and Sagittarius arms. The Sun completes one orbit around the galactic centre approximately every 225–250 million years, a period sometimes called a galactic year.

The galactic centre hosts Sagittarius A*, the supermassive black hole of about 4 million solar masses. Orbits of stars near the galactic centre (especially the star S2) have been tracked for decades; their orbital motions reveal the mass and position of Sagittarius A* with extraordinary precision. Reinhard Genzel and Andrea Ghez shared the 2020 Nobel Prize in Physics for this work.

The Milky Way is surrounded by a spherical halo of older stars, globular clusters (dense balls of 10^4 to 10^6 stars), and an enormous dark matter halo extending far beyond the visible disc. The dark matter halo has a mass several times that of all the visible material in the galaxy.

52.9 Redshift and Cosmic Expansion

In 1929, Edwin Hubble announced a remarkable discovery: almost all galaxies outside the Local Group are moving away from us, and the more distant they are, the faster they recede. This observation is based on the redshift of galaxy spectra.

The Doppler effect (Chapter 22) causes the wavelength of light from a receding source to be stretched to longer (redder) wavelengths. By measuring the redshift z of a galaxy's spectral lines — comparing the observed wavelength to the laboratory wavelength — astronomers can determine the galaxy's recession velocity.

$$z = (\lambda_{obs} - \lambda_{emit}) / \lambda_{emit}$$

Redshift z. For small z, the recession velocity v ≈ zc. For cosmological redshifts, z results not from the Doppler effect of motion through space but from the stretching of space itself: photons are redshifted because the universe has expanded during their travel.

$$v = H_0 d$$

Hubble's Law. The recession velocity v of a galaxy (km/s) is proportional to its distance d (Mpc). H_0 is the Hubble constant ≈ 70 km/s/Mpc (current best value), meaning a galaxy 1 Mpc away recedes at about 70 km/s; one 100 Mpc away recedes at 7000 km/s.

Hubble's law is not because galaxies are moving through space away from us — it is because space itself is expanding. Every region of space is stretching, carrying galaxies apart like raisins in rising bread dough. This expansion has no centre: every galaxy sees every other galaxy receding, consistent with a universe that is expanding everywhere simultaneously.

Looking at very distant galaxies is looking back in time. The most distant galaxies observed have redshifts of $z > 10$, meaning their light was emitted when the universe was less than 5% of its current age. The expansion of space since then has stretched this light, greatly redshifting it into the infrared. The James Webb Space Telescope (launched 2021) is designed to observe these early galaxies.

52.10 The Big Bang Model

If the universe is currently expanding, then in the past it was smaller, denser, and hotter. Extrapolating backward in time leads to a moment when all the matter and energy in the observable universe was compressed into an extremely hot, dense state — the Big Bang singularity — approximately 13.8 billion years ago. The Big Bang model is the cosmological standard model, supported by multiple independent lines of evidence.

The early universe timeline, in brief:

Step 1: **Planck epoch ($t < 10^{-43}$ s):** Physics as currently known does not apply; quantum gravity effects dominate. We have no confirmed theory of this era.

Step 2: **Quark epoch ($t \sim 10^{-6}$ s):** Quarks and gluons condense into protons and neutrons. The universe is a quark-gluon plasma.

Step 3: **Nucleosynthesis ($t \sim 3$ min):** Protons and neutrons fuse to form helium-4 (about 25% by mass), deuterium, lithium, and a small amount of heavier elements. About 75% of mass remains as hydrogen. These ratios match observations precisely.

Step 4: **Recombination ($t \sim 380\,000$ yr):** The universe cools enough for electrons to combine with nuclei to form neutral atoms. Space becomes transparent to photons for the first time. The released photons are the CMB (Section 52.11).

Step 5: **First stars ($t \sim 200$ million yr):** The first stars ignite, beginning the reionisation of the universe and the synthesis of heavier elements.

Step 6: **Galaxy formation ($t \sim 500$ million yr):** Galaxies begin to form around dark matter density fluctuations.

Step 7: **Present ($t \sim 13.8$ billion yr):** The universe has its current large-scale structure. Dark energy dominates the energy budget, driving accelerated expansion.

The Big Bang was not an explosion of matter into pre-existing empty space. It was an expansion of space itself. There was no pre-existing space, and there is no centre of the explosion. Every location in the universe is equally a starting point.

Common Mistake: The Big Bang model does not describe or explain what happened "before" the Big Bang, or why the universe began in the state it did. These remain open questions at the frontier of theoretical physics and cosmology. The Big Bang model is extraordinarily well-supported as a description of what happened after the initial moment, but it is not a complete theory of origins.

52.11 Cosmic Microwave Background Radiation

The Cosmic Microwave Background (CMB) is the afterglow of the Big Bang — thermal radiation left over from the recombination epoch, about 380 000 years after the Big Bang. It fills the entire universe uniformly and corresponds to a blackbody temperature of 2.725 K. It was discovered accidentally in 1965 by Arno Penzias and Robert Wilson at Bell Labs (who received the Nobel Prize in 1978).

The CMB is the most perfect blackbody spectrum ever measured in nature. Its isotropy (the temperature is nearly the same in every direction) and its tiny temperature fluctuations (about 1 part in 10^5) provide a snapshot of the universe at 380 000 years of age. The pattern of these fluctuations encodes the density variations that eventually grew, under gravity, into the galaxies and galaxy clusters we observe today.

Detailed maps of CMB temperature fluctuations by the COBE (1989), WMAP (2001), and Planck (2009) satellites have measured the geometry, composition, and age of the universe with extraordinary precision. These maps confirm that the universe is spatially flat (to within 0.4%), that it is 13.8 billion years old, that it is composed of approximately 5% ordinary matter, 27% dark matter, and 68% dark energy, and that the fluctuations in the CMB are consistent with a period of very rapid early expansion called inflation.

> **Key Point:** The CMB also provides confirmation that the early universe was extremely hot and dense: the photons of the CMB have been travelling to us since recombination and have been redshifted by a factor of about 1100 from the visible/near-infrared to the microwave portion of the spectrum. This factor of 1100 means the universe has expanded by that factor in linear scale since recombination.

52.12 Dark Matter and Dark Energy

The standard cosmological model requires that only about 5% of the total energy content of the universe is ordinary (baryonic) matter — the protons, neutrons, and electrons that make up everything we can directly observe. The remaining 95% consists of two mysterious components: dark matter (about 27%) and dark energy (about 68%).

Dark Matter

Dark matter is matter that does not interact with light (it neither emits nor absorbs electromagnetic radiation) but does have gravitational effects. Evidence for dark matter is overwhelming and comes from multiple independent observations. Galaxy rotation curves (the orbital speeds of stars in galaxies remain too high at large radii to be explained by visible matter alone), gravitational lensing patterns, the structure of galaxy clusters, the growth of large-scale structure, and the CMB power spectrum all require the existence of additional mass that is not visible.

The nature of dark matter remains unknown. The leading candidates are Weakly Interacting Massive Particles (WIMPs) — hypothetical heavy particles that interact only through gravity and the weak nuclear force — and axions — extremely light hypothetical particles. Despite extensive searches using underground detectors, collider experiments, and astrophysical observations, no dark matter particle has been directly detected as of 2025.

Dark Energy

In 1998, two independent teams studying Type Ia supernovae (which serve as standard candles for distance measurement) discovered that the expansion of the universe is accelerating, not decelerating. This was entirely unexpected: gravity should be slowing the expansion. The cause of the accelerated expansion is called dark energy.

Dark energy is most simply modelled as the cosmological constant Λ (lambda) — a uniform energy density of empty space — originally introduced by Einstein in 1917 (for the wrong reasons) and later abandoned. Its physical origin is unknown. The problem of explaining why the vacuum has the particular energy density it does (measured to be about 10^{-29} J/m³) is one of the deepest unsolved problems in theoretical physics.

Component	Fraction of Universe	What We Know	Nature
Ordinary matter	~ 5%	Protons, neutrons, electrons; all visible objects	Well understood
Neutrinos	~ 0.5%	Extremely light, travel near c, rarely interact	Understood (mostly)
Dark matter	~ 27%	Has gravity, no EM interaction, unknown particle	Unknown
Dark energy	~ 68%	Causes accelerated expansion; fills all space	Unknown

Table 52.4 *The energy-matter content of the universe based on CMB and large-scale structure observations.*

52.13 Observational Astronomy and Evidence

Modern cosmology is built on observation, not speculation. Multiple independent observational techniques provide mutually consistent evidence for the standard model of cosmology.

Spectroscopy

The absorption and emission spectra of stars and galaxies reveal their chemical composition, temperature, and radial velocity. Nearly all elements found in stellar spectra are those predicted by stellar nucleosynthesis models. The hydrogen-to-helium ratio in the universe (about 75:25 by mass) matches Big Bang nucleosynthesis predictions precisely.

Standard Candles

Certain objects have known intrinsic luminosity, making them cosmic distance indicators. Type Ia supernovae all reach approximately the same peak brightness (because they result from white dwarfs exceeding the Chandrasekhar mass limit of about 1.4 solar masses). Cepheid variable stars pulsate with periods related to their intrinsic luminosity. These standard candles allow distance measurements to billions of light-years.

Gravitational Lensing

Massive objects bend light paths (general relativity). The pattern of lensing around galaxy clusters reveals the distribution of total mass — both visible and dark. The Bullet Cluster (a pair of galaxy clusters that passed through each other) shows dark matter and visible matter separated after the collision, providing some of the most compelling evidence for dark matter as a distinct, non-baryonic component.

Gravitational Wave Astronomy

Since 2015, LIGO and Virgo have detected dozens of gravitational wave events from merging black holes and neutron stars. Neutron star mergers (kilonova events) have been directly observed to produce heavy elements through the r-process, confirming that events like these are the astrophysical site of heavy element synthesis.

The consistency among spectroscopy, redshift measurements, the CMB, gravitational lensing, and gravitational wave observations provides extraordinary confidence in the broad outlines of modern cosmology, even as many details remain active research areas.

52.14 Common Misunderstandings About Cosmology

- The Big Bang happened at a specific location in space — Incorrect. The Big Bang was an expansion of space itself, occurring everywhere simultaneously. There is no central point from which the universe expanded. Every location is equally a starting point.
- Galaxies are moving through space away from us — Partially misleading. For nearby galaxies with significant peculiar velocities, motion through space matters. But for most galaxies, especially distant ones, the dominant cause of recession is the expansion of space itself stretching the distances between galaxies. Galaxies are largely stationary in their local space; it is the space between them that is expanding.

- We are at the centre of the universe because everything moves away from us — If the universe is expanding uniformly (Hubble's Law), every galaxy sees every other galaxy receding. There is no special centre. An observer in any other galaxy would see the same pattern.
- Dark matter is ordinary matter that is simply dark — Dark matter is not dust, gas, dim stars, or black holes (though these contribute negligibly). Dark matter is a new type of particle that does not interact with the electromagnetic force at all and has never been directly detected in a laboratory.
- The Big Bang and stellar evolution contradict "creation" — This is a philosophical or theological statement, not a scientific one. Cosmology describes what happened; it does not address why there is something rather than nothing, or questions of ultimate origins that lie outside the scientific method.
- The universe is 13.8 billion years old, so we cannot see anything farther than 13.8 billion light-years — Incorrect. Because space has been expanding since the Big Bang, the most distant objects we can observe are about 46 billion light-years away today (even though their light took only 13.8 billion years to reach us, the objects have moved much farther away due to cosmic expansion during that time).
- The Sun will explode as a supernova — The Sun is too low in mass to undergo a supernova. When its hydrogen is exhausted in about 5 billion years, it will expand into a red giant, shed its outer layers as a planetary nebula, and leave behind a white dwarf. Only stars more massive than about 8 solar masses end in supernovae.

Common Mistake: Dark matter and dark energy are named for what they are not (visible, like ordinary matter) and for a functional description (energy causing expansion), respectively. Neither term implies that these components are mysterious or non-scientific — both are precisely constrained by quantitative observations and are essential components of the standard cosmological model.

Chapter Summary

The same physical laws governing atoms and forces on Earth explain the formation, life, and death of stars, the structure of galaxies, and the expansion and history of the universe.

- Gravity is the dominant force at cosmic scales: always attractive, long-range, cumulative. It forms stars, galaxies, clusters, and the cosmic web.

- Stars form from collapsing nebulae; fusion ignites when core temperature reaches ~10 million K. Hydrostatic equilibrium balances gravity and radiation pressure.
- The proton-proton chain fuses hydrogen into helium in main-sequence stars; heavier elements are built up to iron by successive fusion stages in massive stars.
- A star's fate depends on mass: low-mass stars become white dwarfs; massive stars end as neutron stars or black holes via supernova.
- Supernovae synthesise elements heavier than iron and distribute them into the interstellar medium.
- Hubble's Law ($v = H_0 d$) shows the universe is expanding; looking further is looking further back in time.
- The Big Bang model describes the universe's origin 13.8 billion years ago as an expansion from a hot, dense state. Big Bang nucleosynthesis accounts for the observed H/He ratio.
- The CMB is the thermal afterglow of recombination (380 000 years after the Big Bang); its temperature fluctuations encode the seeds of all cosmic structure.
- The universe is approximately 5% ordinary matter, 27% dark matter, and 68% dark energy. Dark matter's nature is unknown; dark energy causes accelerated expansion.
- Multiple independent observational techniques (spectroscopy, CMB, gravitational lensing, standard candles, gravitational waves) converge on a consistent cosmological model.

Key Terms

nebula: A cloud of gas and dust in interstellar space; the birthplace of stars.

protostar: A collapsing mass of gas in the early stage of star formation, before fusion ignites.

main sequence: The stable phase of a star's life during which hydrogen is fused to helium in the core; lasts millions to trillions of years depending on mass.

red giant: A star that has exhausted core hydrogen and expanded dramatically; outer layers are cool and red.

white dwarf: The collapsed remnant of a low-mass star; about Earth-sized; supported by electron degeneracy pressure; no longer fusing fuel.

supernova: The explosive death of a massive star when its iron core collapses; briefly outshines the entire galaxy; synthesises and disperses heavy elements.

neutron star: The collapsed remnant of a supernova from a star of 8–20 solar masses; ~20 km in diameter; supported by neutron degeneracy pressure; density ~ nuclear density.

black hole: A region of spacetime where gravity is so strong that not even light can escape; formed from collapsing stellar cores or in galaxy centres.

event horizon: The boundary of a black hole within which the escape velocity exceeds c; no information can escape.

galaxy: A gravitationally bound system of stars, gas, dust, and dark matter, typically containing millions to trillions of stars.

Hubble's Law: $v = H_0 d$; the recession velocity of a galaxy is proportional to its distance; evidence for cosmic expansion.

redshift (z): The stretching of light wavelengths from a receding source; for cosmological redshifts, caused by the expansion of space.

Big Bang: The model describing the origin of the universe 13.8 billion years ago as an expansion from an extremely hot, dense state.

Big Bang nucleosynthesis: The production of hydrogen, helium, and lithium in the first few minutes after the Big Bang; predicts the observed H/He ratio.

Cosmic Microwave Background (CMB): The thermal radiation left over from the recombination epoch (~380 000 years after the Big Bang); temperature 2.725 K; provides a snapshot of the early universe.

dark matter: A form of matter that does not interact electromagnetically but has gravitational effects; constitutes ~27% of the universe's energy content.

dark energy: A form of energy filling all space, causing the accelerated expansion of the universe; constitutes ~68% of the universe's energy content; nature unknown.

stellar nucleosynthesis: The formation of chemical elements heavier than lithium inside stars and in supernova explosions.

pulsar: A rapidly rotating neutron star emitting beams of electromagnetic radiation that appear as regular pulses to observers.

Hubble constant (H_0): The proportionality constant in Hubble's Law; current best value $\approx$70 km/s/Mpc; quantifies the rate of cosmic expansion.

Concept Review Questions

Answer in complete sentences.

1. Why is gravity the dominant force at cosmic scales when it is the weakest of the four fundamental forces at subatomic scales?
2. Describe the sequence of events from a gas cloud to a main-sequence star. What physical process provides the energy that halts the collapse?
3. Explain why a more massive star lives a shorter life than a less massive star, even though it has more fuel.
4. Trace the life cycle of a star like the Sun from formation to its eventual white dwarf stage. What happens to its outer layers?
5. What is the proton-proton chain? Why does nuclear fusion release energy for elements up to iron but not for elements heavier than iron?
6. Explain how Hubble's observations of galaxy recession velocities lead to the conclusion that the universe is expanding.
7. What is the cosmic microwave background? When was it produced, and what does its existence tell us about the early universe?
8. Distinguish dark matter from dark energy. What evidence supports the existence of each?
9. Why does looking at a distant galaxy mean looking back in time? What is the most distant epoch we can observe directly, and what observational tool lets us see it?
10. List four independent observational lines of evidence that support the Big Bang model.

Stellar Life Cycle Exercises

11. A star forms with 0.5 solar masses. (a) What colour will it be on the main sequence? (b) Approximately how long will it live? (c) What will it become when it dies? Will it undergo a supernova?
12. A star has 20 solar masses. (a) What colour is it on the main sequence? (b) Approximately how long will it live? (c) What stages of fusion will occur in its core? (d) What will be the final compact remnant?
13. Explain why iron-56 is the end point of stellar fusion. What happens to the star's core when an iron core forms?
14. A white dwarf has a mass of 0.8 solar masses and a radius of 5000 km. (a) Calculate its average density. (b) Compare it to the density of water (1000 kg/m^3).

(c) What keeps the white dwarf from collapsing further? (Solar mass = 2×10^{30} kg)

15. A neutron star has a mass of 1.4 solar masses and a radius of 10 km. (a) Calculate its average density. (b) How does this compare to nuclear density (~ 2×10^{17} kg/m^3)?

Galaxy and Redshift Analysis

16. Galaxy A has a redshift z = 0.05. (a) Using $v \approx zc$, find its recession velocity. (b) Using H_0 = 70 km/s/Mpc, find its distance in Mpc and in light-years. ($c = 3 \times 10^5$ km/s)
17. A galaxy is observed with a recession velocity of 14 000 km/s. (a) Using Hubble's Law with H_0 = 70 km/s/Mpc, find its distance in Mpc. (b) Convert to light-years. (1 Mpc $\approx 3.26 \times 10^6$ ly)
18. A hydrogen spectral line has a laboratory wavelength of 656 nm (the H-alpha line). In the spectrum of a distant galaxy, this line is observed at 787 nm. (a) Calculate the redshift z. (b) Calculate the recession velocity (in km/s). (c) Estimate the distance using H_0 = 70 km/s/Mpc.
19. Explain why the most distant observable galaxies appear so much redder than nearby galaxies. What has happened to the photons during their travel?

Cosmology Concept Questions

20. Big Bang nucleosynthesis predicts that the universe should be about 75% hydrogen and 25% helium by mass after the first few minutes. Explain in terms of the conditions of the early universe why heavier elements were not also synthesised in significant quantities at that time.
21. The CMB temperature is 2.725 K today. At the time of recombination ($z \approx 1100$), what was the CMB temperature? (Hint: temperature scales as $T \propto (1 + z)$.)
22. If dark energy causes the universe's expansion to accelerate, what does this imply for the long-term fate of the universe? (Qualitative description.)

23. Describe the observational evidence from the Bullet Cluster that separates dark matter from ordinary matter. Why is this considered compelling evidence that dark matter is a distinct, non-baryonic component?

Applied Astronomy Problems

24. The Sun's luminosity is 3.85×10^{26} W and it fuses hydrogen to helium. In this process, 0.7% of the hydrogen mass is converted to energy. (a) Using $P = \Delta E/\Delta t$ and $E = \Delta mc^2$, calculate the rate at which the Sun converts mass to energy (kg/s). (b) The Sun's mass is 2.0×10^{30} kg. Assuming a constant luminosity, how long could it shine before running out of mass entirely? (This overestimates its life; in reality, only the core hydrogen is accessible.)
25. A Type Ia supernova reaches a peak luminosity of about 1.0×10^{37} W. A supernova is observed with a peak apparent flux of 1.5×10^{-17} W/m². (a) Using the inverse square law ($F = L / (4\pi d^2)$), calculate its distance d. (b) Convert to light-years. ($c = 3 \times 10^8$ m/s; 1 ly $\approx 9.46 \times 10^{15}$ m)
26. The escape velocity from the event horizon of a black hole is c. Using $v_{esc} = \sqrt{2GM/r} = c$, derive the Schwarzschild radius r_s (the radius of the event horizon) in terms of M and fundamental constants. Then calculate r_s for (a) a 10 solar mass black hole and (b) a 4×10^6 solar mass black hole (Sagittarius A*). ($G = 6.67 \times 10^{-11}$ N·m²/kg²; solar mass = 2×10^{30} kg)

Chapter 52 Checkpoint

Before moving on to Part X, confirm you can do each of the following.

I can...	See Section
Explain why gravity dominates at cosmic scales despite being the weakest force	52.2
Describe how stars form from nebulae and the role of hydrostatic equilibrium	52.3
Explain the proton-proton chain and why stellar fusion ends at iron	52.4

Trace the life cycle of a low-mass and a high-mass star to their respective remnants	52.5
Describe what a supernova produces and why it is critical for element synthesis	52.6
Classify galaxy types and describe the structure of the Milky Way	52.7–52.8
State Hubble’s Law and explain what galaxy redshifts tell us about the universe	52.9
Describe the key stages of the Big Bang model and Big Bang nucleosynthesis	52.10
Explain what the CMB is, when it was produced, and what it tells us	52.11
Describe the evidence for dark matter and dark energy and their proportions	52.12
Name four observational techniques used to support the cosmological model	52.13
Identify and correct common misunderstandings about the Big Bang and cosmology	52.14

Continue to Chapter 53: Physics in Engineering →